



# Vanderbilt Genome Editing Resource

## GENOTYPING PROTOCOL: Rr606<sup>em1Mgn</sup>

This strain may be genotyped using either or both of the following two PCR assays. The two deletions are within about 150 kb of each other and are not expected to segregate by recombining with the WT allele as they are within genetic linkage distance. That being said, running both assays to verify both deletions is the preferred genotyping method.

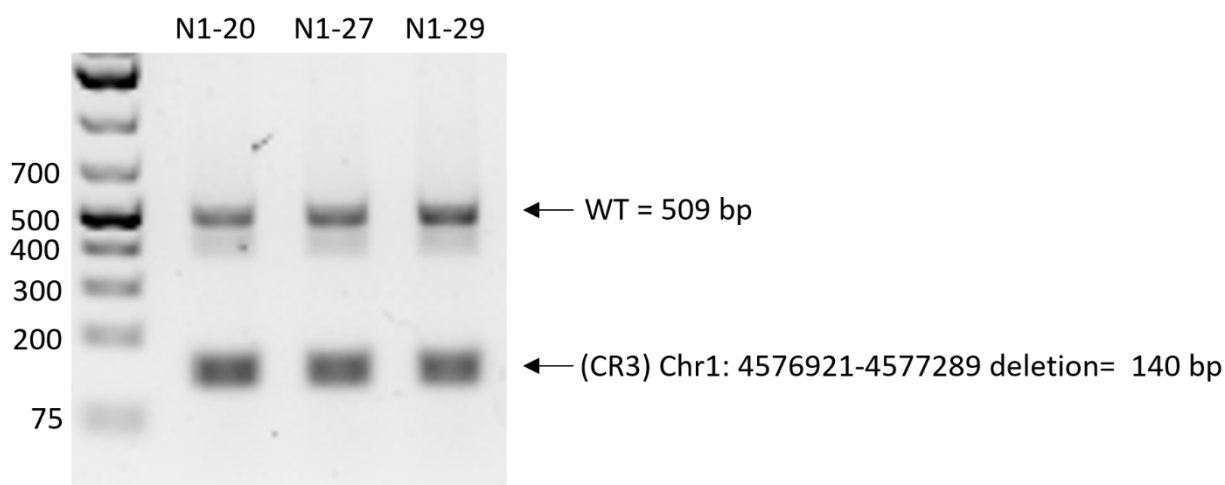
### PCR assay for (CR3) Chr1: 4576921-4577289 deletion

| Name             | Primer sequence              | Primer description |
|------------------|------------------------------|--------------------|
| <b>GtCR3-Fwd</b> | CTGGAGCTTCATTGCATCGTTATAAGAG | WT/mutant Fwd      |
| <b>GtCR3-Rev</b> | GTGAGTCCTACTGCAAAGTGACTAC    | WT/mutant Rev      |

#### Expected PCR products:

WT = 509 bp

(CR3) Chr1: 4576921-4577289 deletion= 140 bp



| Polymerase            | EconoTaq PLUS GREEN 2x Master Mix, Lucigen Catalog #3033-1                                                                                                                                                                                                                                                                 |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 20 µl reaction volume | 10 µl, EconoTaq PLUS GREEN 2x Master Mix<br>7 µl water<br>1.0 µl 10 µM site-specific Rev<br>1.0 µl 10 µM site-specific Rev<br>1 µl purified genomic DNA or lysate (50-250 ng)                                                                                                                                              |
| Cycling Conditions    | <ol style="list-style-type: none"> <li>1. Initial denaturation, 94°C, 2 min</li> <li>2. Denaturation, 94°C, 30 seconds</li> <li>3. Annealing, 65°C, 30 seconds</li> <li>4. Extension, 72°C, 45 seconds, Go to 2 for 40 cycles.</li> <li>5. Final extension, 72°C, 5 minutes</li> <li>6. Hold, 4°C, indefinitely</li> </ol> |



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## GENOTYPING PROTOCOL: Rr606<sup>em1Mgn</sup>

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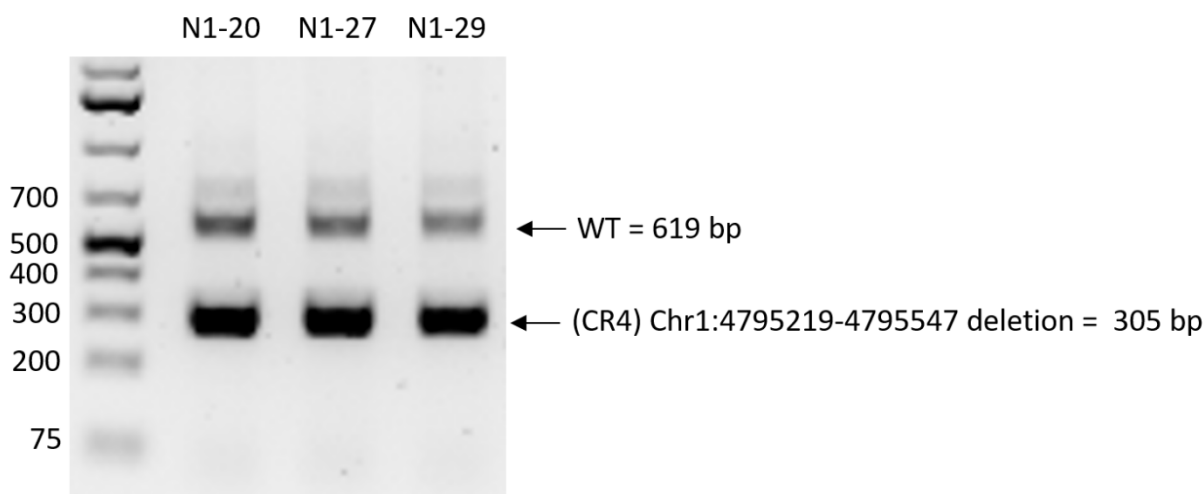
### PCR assay for (CR4) Chr1:4795219-4795547 deletion

| Name           | Primer sequence         | Primer description |
|----------------|-------------------------|--------------------|
| Sox17-CR4-Fwd1 | CTTTCATTCTAGTGAGTGGATCC | WT/mutant Fwd      |
| Sox17-CR4-Rev1 | CATAAGACCACACTGAACAAACC | WT/mutant Rev      |

### Expected PCR products:

WT = 619 bp

(CR4) Chr1:4795219-4795547 deletion = 305 bp



| Polymerase            | EconoTaq PLUS GREEN 2x Master Mix, Lucigen Catalog #3033-1                                                                                                                                                                                                                                                                 |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 20 µl reaction volume | 10 µl, EconoTaq PLUS GREEN 2x Master Mix<br>7 µl water<br>1.0 µl 10 µM site-specific Rev<br>1.0 µl 10 µM site-specific Rev<br>1 µl purified genomic DNA or lysate (50-250 ng)                                                                                                                                              |
| Cycling Conditions    | <ol style="list-style-type: none"> <li>1. Initial denaturation, 94°C, 2 min</li> <li>2. Denaturation, 94°C, 30 seconds</li> <li>3. Annealing, 61°C, 30 seconds</li> <li>4. Extension, 72°C, 45 seconds, Go to 2 for 40 cycles.</li> <li>5. Final extension, 72°C, 5 minutes</li> <li>6. Hold, 4°C, indefinitely</li> </ol> |